

Nonequilibrium Models of Strategic Thinking

John Van Huyck

Overview

- Strategic game theory uses three distinct concepts to make precise predictions of how people will, or should, interact strategically:
 - ▶ Formation of beliefs based on analysis of what others might do (strategic thinking);
 - ▶ choosing a best response given those beliefs (optimization);
 - ▶ and adjustment of best responses and beliefs until they are mutually consistent (equilibrium).
- Most applications of game theory assume equilibrium
 - ▶ learning has converged to a stochastic steady state
 - ▶ equilibrium approximates people's strategic thinking
- Quantal Response Equilibrium weakens the optimization assumption, but maintains equilibrium strategic thinking.
- Level- k / Cognitive Hierarchy Models emphasize the lack of mutual consistency and heterogeneity in initial behavior

Laboratory Evidence on Mutual Consistency

- Recall the p-beauty contest, see Nagel (1995), Ho, Camerer, and Weigelt (1998), Nagel and Duffy (1999).
- Van Huyck, Battalio, and Beil (1990) illicit beliefs that were heterogenous but most people gave a best response to those beliefs.
- Stahl and Wilson (1995) formally reject the mutual consistency condition.
- Camerer (2003) emphasizes that even when we observe subjects obeying iterated dominance in specific situations it may be that they are using a decision rule that does not satisfy dominance in general.

Nonequilibrium Approach

- Nonequilibrium approach is appropriate as
 - ▶ model of initial behavior
 - ▶ allows better prediction of history-dependent limiting outcomes
- A form of bounded rationality, which may be a more accurate description of behavior,
 - ▶ avoids fixed-point reasoning
 - ▶ assumes limited iterative best response reasoning

Level- k / Cognitive Hierarchy Models

- Initially people's strategic thinking is heterogeneous
 - ▶ simple player “types” (rules of thumb rather than private information)
 - ★ anchor beliefs in an “instinctive” reaction to a game
 - ★ engage in a small number of rounds of iterative best responses
 - ▶ types are drawn from a population distribution concentrated on one to three best-response iterations

Stahl (GEB,1993) Evolution of Smart-n Players

- Models the evolution of strategic intelligence
- player types are drawn from a hierarchy of "smartness" analogous to the levels of iterated rationalizability.
- Nonrationalizable strategies die out, but when higher levels of smartness incur maintenance costs, being right is always as good as being smart.
- Moreover, if a manifest way to play emerges, then dumb players never die out, while smarter players with positive maintenance costs vanish.
- These results call to question the standard game-theoretic assumption of super-intelligent players.
- Results used to justify truncating the hierarchy at *level-2* in Stahl & Wilson (GEB, 1995)

Stahl & Wilson (GEB, 1995) On Players' Models of Other Players

- Develops and tests a theory of human behavior in 3x3 symmetric games.
- Five boundedly rational types distinguished by
 - ▶ their model of other players
 - ▶ their ability to identify optimal choices given their priors
- They reject a rational expectations type in favor of the boundedly rational types
- Claim theory provides a stepping stone toward a descriptive and prescriptive game theory

Stahl & Wilson (GEB,1995) Model

- Focus on initial behavior rather than learned behavior
 - ▶ initial responses are difficult to predict
 - ▶ exhibit more diversity
 - ▶ are crucial to whatever learning follows
 - ▶ reveals “model of the world” players bring to new strategic settings
- Consider symmetric games with two players and three strategies.
 - ▶ Let $j \in \{1, 2, 3\}$ denote a strategy (row)
 - ▶ Let U_i denote the 3×3 matrix of expected utility payoffs for games $i \in \{1, 2, 3, \dots, 12\}$
 - ▶ Let U_{ijk} denote row's payoff in game i when row selects j and column selects k
 - ▶ Let $p \in \Delta \equiv \{x \in \mathbb{R}_+^3 \mid \sum_j x_j = 1\}$, the beliefs about the frequency of choices by others
 - ▶ Let p_0 denote the uniform prior

Stahl & Wilson (GEB, 1995) Model continued

- For game i , $U_i p_i$ is the 3×1 vector of expected utility payoffs
- $U_{ij} p_i$ is the expected payoff of strategy j facing a population whose distribution across strategies is p_i
- Let $b(p_0)$ denote the best response to the uniform distribution
- Let $b^2(p_0) \equiv b(b(p_0))$
- Let $p_i^{NE} \in \Delta$ denote the Nash Equilibrium strategy vector for game i
- Define the following two-parameter family of priors

$$q_{ij}(\mu, \varepsilon) \equiv \varepsilon \frac{e^{\mu U_{ij} p_i}}{\sum_k e^{\mu U_{ik} p_i}} + (1 - \varepsilon) p_{ij}^{NE}$$

$$y_{ij}(\mu, \varepsilon) \equiv U_{ij} q_i(\mu, \varepsilon)$$

- Recall the two-parameter family of priors

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- If $\varepsilon = 0$, then we have the Nash prior
- If $\varepsilon = 1$ and $\mu = 0$, then we have the *level-1* prior
- If $\varepsilon = 1$ and $\mu > 0$, then we have the *level-2* prior

Stahl & Wilson's archetypes

- Recall the two-parameter family of priors



$$q_{ij}(\mu, \varepsilon) \equiv \varepsilon \frac{e^{\mu U_{ij} p_i}}{\sum_k e^{\mu U_{ik} p_i}} + (1 - \varepsilon) p_{ij}^{NE}$$

$$y_{ij}(\mu, \varepsilon) \equiv U_{ij} q_i(\mu, \varepsilon)$$

- Players respond to their priors using a logit response function

$$p_{ij}(\gamma, \mu, \varepsilon) \equiv \frac{e^{\gamma y_{ij}(\mu, \varepsilon)}}{\sum_k e^{\gamma y_{ik}(\mu, \varepsilon)}}$$

- Divide the $(\gamma, \mu, \varepsilon)$ space into five “archetypes”
 - ▶ If $\gamma = 0$, then we have *level-0* type: uniform play
 - ▶ If $\gamma > 0$, $\mu = 0$ and $\varepsilon = 1$, then we have *level-1* type: logit response to uniform prior
 - ▶ if $\gamma > 0$, $\mu > 0$ and $\varepsilon = 1$, then we have *level-2* type: logit response to logit response to uniform prior
 - ▶ if $\gamma > 0$ and $\varepsilon = 0$, then we have a *naive Nash* type: logit response to Nash prior
 - ▶ if $\gamma > 0.1$ and $\varepsilon \in (0, 1)$, then we have a *worldly* type

Stahl & Wilson's Multimodal Model

- Let $s(h, i) \in \{1, 2, 3\}$ denote the strategy chosen by participant h in game i
- Let $s^h \equiv \{s(h, i), i = 1, \dots, 12\}$ denote the choices of participant h
- *Assuming that a participant's type is fixed for all games*
- the probability of participant h 's joint choices conditional on being level- l is

$$p_l^h(\gamma_l, \mu_l, \varepsilon_l) \equiv \prod_i p_{is(h,i)}(\gamma_l, \mu_l, \varepsilon_l), \text{ for } l = 1, 2, 3, 4$$

- Excluding the RE type, the model has five types: Let β denote the complete vector of all type parameters
- The ex ante likelihood of participant h 's joint choices is given by

$$L(s^h | \beta) = \alpha_0 p_0 + \sum_{l=1}^4 \alpha_l p_l^h(\gamma_l, \mu_l, \varepsilon_l)$$

- The log-likelihood of the observed sample is

$$\mathcal{L} \equiv \sum_h \log[L(s^h | \beta)]$$

Stahl & Wilson's Experimental Design

- Requirement of common knowledge understanding suggests symmetric 3×3 games
 - ▶ simple to understand
 - ▶ yet rich enough to separate the various player types
- Participants are always a row player and choose one strategy for each of 12 games, that is, play each game once
- No feedback while playing the games
- Each participant choices were matched against all other participants, “playing against the field.”
- Numbers in the payoff table were the chance of winning \$2.00
- Instructions followed Cooper et al. (1990)
- Actual monetary winnings ranged from \$4.00 to \$18.00 with an average of \$11.63
- Sessions last about 75 minutes
- Participants were fourth and fifth year undergraduate accounting and finance majors at UT
- Three sessions of 14, 22, and 12 participants respectively

Stahl & Wilson's games

Table : Stahl & Wilson's games, model predictions, and actual frequencies

	Game			Actual	Predicted	Types
Game 1	T	M	B			
T	25	30	100	0.15	0.19	$b(p_0)$
M	40	45	65	0.83	0.76	Nash, $b^2(p_0)$
B	31	0	40	0.02	0.06	D1
Game 2	T	M	B	Actual	Predicted	Types
T	75	40	45			
M	70	15	100	0.25	0.28	$b(p_0)$
B	70	60	0	0.13	0.21	$b^2(p_0)$

- See Camerer (2003) p.244-24 for all 12 games



Stahl & Wilson's results

Table : Parameter Estimates For Mixture Model Without RE Types

	Estimate	Std. Dev.
γ_1	0.22	0.04
μ_2	0.46	0.06
γ_2	3.08	0.57
γ_3	4.99	0.94
μ_4	0.06	0.01
ε_4	0.44	0.08
γ_4	0.33	0.05

	Estimate	Std. Dev.
α_0	0.17	0.06
α_1	0.21	0.06
α_2	0.02	0.02
α_3	0.17	0.06
α_4	0.43	0.08
$\mathcal{L}$	-442.73	

Discussion

- Average payoff ranged from 43 for *level-0* to 47 for *level-1*, *naive Nash*, and *worldly* types.
 - ▶ Estimates are not significantly different from each other
 - ▶ Selection pressure against boundedly rational types may be weak, see Stahl (1993)
- Crawford et al. (2010) argue on apriori grounds that the *worldly* type should be excluded
 - ▶ depends on estimated parameters
 - ▶ thus, implicitly assumes that subjects have prior understanding of others' responses
- Stahl and Wilson (1994) found large numbers of *level-1* and *level-2* types, as do Costa-Gomes, Crawford, and Broseta (2001)
- This parameter estimate instability reflects a common trade-off
 - ▶ data analysis rests heavily on small-sample econometrics
 - ▶ danger of specification bias and correspondingly fragile estimates
- Costa-Gomes and Crawford (2006), which lets the data speak for themselves, reject *worldly* type in favor of *level-2* type

Haruvy and Stahl (2003) Level-n Bounded Rationality and Dominated Strategies

- Stylized Fact: subjects generally avoid dominated strategies, but seldom iteratively eliminate dominated strategies.
 - ▶ beauty contest games: Nagel (1995); Stahl (1996)
 - ▶ two-iteration dominance solvable symmetric strategic form games: Stahl and Wilson (1994, 1995)
 - ▶ two- and three-iteration dominance solvable games: Costa-Gomes et al. (2001)
 - ▶ nine-iteration dominance solvable games: Sefton and Yavas (1996); Katok et al. (2002)
- Is this finding and the Stahl and Wilson (1995) model robust to the presence of obviously dominated strategies?

Obviously Dominated Strategies

- Some dominated strategies are less obvious than others
 - ▶ Strategies dominated by a mixed strategy
 - ▶ Dominated strategies with payoffs above the maximin payoff level
- Stahl and Wilson (1995) level- n bounded rationality model may not be robust obviously dominated strategies. For example,

	<i>Top</i>	<i>Middle</i>	<i>Bottom</i>
<i>Top</i>	0	0	0
<i>Middle</i>	10	90	10
<i>Bottom</i>	100	5	20

- If *level-0* types avoid obviously dominated strategies, then *level-1* types should not assign probability $1/3$ to *Top*
 - ▶ *level-1* types with SW95 beliefs, play *Bottom* but if *level-0* types avoid *Top* then the 100 payout from (*Bottom*, *Top*) will not be realized.
 - ▶ *level-1* types that recognize that *level-0* types will avoid obviously dominated strategies play *Middle* instead

Formalizing Obviously Dominated Strategies

- Let $z_j \equiv \min \{0, m_j - M\}$, where m_j is the maximum payoff of strategy j and M is the maximin payoff of the game
- However, paper rejects the formalization
- But finds support for a linear title model of *level-0* types, which puts about 10 percent less than uniform probability on dominated strategies.
- “...the surprisingly small impact obviously dominated strategies had on subjects’ behavior suggests that level- n thinking is very deeply rooted in subjects’ belief formation and approach to strategic games to the point that it can represent a real bias in behavior and generate ‘obviously unreasonable’ beliefs.”
- However, they conjecture that some learning dynamic will drive out play of obviously dominated strategies.
- Obviously dominated strategies can alter the play path and influence long-run behavior.

References

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